RAJALAKSHMI ENGINEERING COLLEGE

Josenah J 251001098-2025-IT

J2

Started on	Thursday, 17 September 2026, 12:03 AM
State	Finished
Completed on	Thursday, 17 September 2026, 12:25 AM
Time taken	21 mins 23 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main(){
    int n;
    scanf("%d",&n);

    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }

    for(int k=1;k<=n;k++){
        int c = 0;
        for(int j=0;j<n;j++){
            if(a[j] == k){
                c++;
            }
        }
        if(c > 1){
            printf("%d",k);
            break;
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Josenah J 251001098-2025-IT

J2

Started on	Thursday, 17 September 2026, 12:29 AM
State	Finished
Completed on	Thursday, 17 September 2026, 12:33 AM
Time taken	4 mins 2 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    int slow = a[0];
    int fast = a[0];

    do
    {
        slow = a[slow];
        fast = a[a[fast]];
    } while (slow != fast);

    slow = a[0];

    while (slow != fast)
    {
        slow = a[slow];
        fast = a[fast];
    }

    printf("%d", slow);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Josenah J 251001098-2025-IT

Started on	Thursday, 17 September 2026, 12:33 AM
State	Finished
Completed on	Thursday, 17 September 2026, 12:40 AM
Time taken	6 mins 55 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int T;
    scanf("%d", &T);

    while (T--)
    {
        int n1, n2;

        scanf("%d", &n1);

        int a[n1];

        for (int i = 0; i < n1; i++)
        {
            scanf("%d", &a[i]);
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Josenah J 251001098-2025-IT

J2

Started on	Thursday, 17 September 2026, 12:41 AM
State	Finished
Completed on	Thursday, 17 September 2026, 12:45 AM
Time taken	4 mins 46 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int T;
    scanf("%d", &T);

    while (T--)
    {
        int n1, n2;

        scanf("%d", &n1);

        int a[n1];
        for (int i = 0; i < n1; i++)
            scanf("%d", &a[i]);

        scanf("%d", &n2);

        int b[n2];
        for (int i = 0; i < n2; i++)
            scanf("%d", &b[i]);
```

```
        int i = 0, j = 0;

        while (i < n1 && j < n2)
        {
            if (a[i] == b[j])
            {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j])
            {
                i++;
            }
            else
            {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Josenah J 251001098-2025-IT

J2

Started on	Tuesday, 6 October 2026, 11:19 AM
State	Finished
Completed on	Tuesday, 6 October 2026, 11:22 AM
Time taken	3 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int A[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    int k;
    scanf("%d", &k);

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {

            if (A[j] - A[i] == k) {
                printf("1\n");
                return 0;
            }
        }
    }

    printf("0\n");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Josenah J 251001098-2025-IT

J2

Started on	Tuesday, 6 October 2026, 11:22 AM
State	Finished
Completed on	Tuesday, 6 October 2026, 11:25 AM
Time taken	2 mins 35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int A[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0;
    int j = 1;

    while (i < n && j < n) {

        if (i == j) {
            j++;
        }
        else if (A[j] - A[i] == k) {
            printf("1\n");
            return 0;
        }
        else if (A[j] - A[i] < k) {
            j++;
        }
        else {
            i++;
        }
    }

    printf("0\n");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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